DK1 day-ahead prices: methods

github.com/jacobazu/dk1-price-forecasting · October 2026

Forecasts of tomorrow's 24 hourly prices in DK1, made every morning before the noon auction and scored once the prices are out.

Inputs	DK1 prices of D−1, D−2, D−3 and D−7, neighbour prices of D−1, Danish holidays, and weather forecasts for DK1 and Germany made 48 hours ahead. All published before the auction (`inputs.csv`).
Models	BMT-X (Kapetanios et al., 2026), LEAR-X (Lago et al., 2021), XGBoost-X (Chen and Guestrin, 2016) and their average, refitted daily on two years, on asinh-transformed prices (Uniejewski et al., 2018). Benchmark: the weekly naive.
Scoring	MAE and rMAE; Diebold–Mariano tests with a Newey–West variance (Diebold and Mariano, 1995; Newey and West, 1987); Giacomini–White (Giacomini and White, 2006); model confidence set (Hansen et al., 2011); Mincer–Zarnowitz (Mincer and Zarnowitz, 1969).
Ranges	Quantiles of past errors with adaptive conformal inference (Gibbs and Candès, 2021); pinball loss, coverage tests, and the Brier score for negative prices.
Battery	A 2 MWh battery planned on the forecast with a mixed-integer linear program; revenue at risk by EWMA-filtered historical simulation (Hull and White, 1998), tested with Kupiec (1995), Christoffersen (1998) and Acerbi and Székely (2014), and compared with the FZ0 loss (Patton et al., 2019).
Rules	Only data published before the auction, and the backtest uses the same inputs. Test rules are written before the test. Changes found in the backtest run live as challengers first.

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